

JITHENDRA PATHIRAJA

+94 76 343 4630 jithendrapathiraja3434@gmail.com

[in linkedin.com/in/jithendra-pathiraja](https://www.linkedin.com/in/jithendra-pathiraja)

Moratuwa, Sri Lanka

EDUCATION

University of Moratuwa

B.Sc. Engineering (Hons.) in Mechanical Engineering; GPA: 3.69/4.00
Dean's List: Semesters 2, 4, and 6

Moratuwa, Sri Lanka
2022 – Present

Maliyadeva College

GCE Advanced Level – Physical Science Stream; Z-score: 2.1052
GCE Ordinary Level (2017): 9A Passes

Kurunegala, Sri Lanka
2012 – 2020

EXPERIENCE

Steering Engineer

Team FalconE Racing

May 2025 – Present
University of Moratuwa

- Designed and developed an improved steering mount between the steering column and steering shaft to increase rigidity and prevent flex under high steering torque.
- Responsible for maintenance, troubleshooting, and continuous optimization of the vehicle's steering system to ensure performance and reliability.

Team Member – Steering

Team FalconE Racing

Aug 2024 – May 2025
University of Moratuwa

- Studied steering system design principles and performed steering torque calculations for the FalconE Formula Student race car.
- Designed and developed a quick jack for the Falcon E2 Formula Student race car.

Trainee Mechanical Engineer

Vega Innovations (Pvt) Ltd

Dec 2024 – Jun 2025
Colombo, Sri Lanka

- Developed a MATLAB/Simulink model to analyze the front suspension dynamics of the Vega Street Bike and evaluate chassis displacement under bump excitation.
- Performed simulations to study the effect of suspension stiffness and damping on ride response and determine critical damping conditions.
- Designed and developed a rear paddock stand for the Vega Street Bike to enable stable lifting during maintenance operations.
- Designed and developed a reusable welding jig for the Vega Street Bike chassis to improve assembly accuracy during fabrication.

ENGINEERING PROJECTS

Design of High-Speed Ambulance Craft

- Designed a high-speed ambulance vessel using regression-based analysis of existing craft data.
- Established relationships between principal dimensions to ensure stability and hydrodynamic performance.
- Selected optimal vessel dimensions based on statistical modeling and design constraints.

Design and Development of a Hydrodynamic Tunnel (Ongoing)

- Designing a laboratory-scale hydrodynamic tunnel to achieve controlled, low-turbulence flow conditions.
- Performing CFD simulations (ANSYS) to analyze flow behavior and pressure distribution.
- Developing CAD models and optimizing key components for improved flow uniformity.

Steering System Optimization – Falcon E2 Formula Student Car

- Designed and developed an improved steering mount to enhance rigidity under high steering loads.
- Developed a custom jig for precise alignment of bearing housings during fabrication.
- Integrated and optimized steering system to improve performance and reliability.

Quick Jack System – Falcon E2 Formula Student Car

- Designed and fabricated a mechanical quick jack system for efficient lifting of the vehicle.
- Selected structural members to ensure strength, stability, and safe operation under load.
- Optimized design for ease of use and reliable load distribution.

Prosthetic Leg Fatigue Testing Mechanism

- Designed a mechanical system to simulate human gait for fatigue testing of prosthetic legs.
- Developed linkage mechanism and loading system to replicate realistic walking conditions.
- Modeled and fabricated prototype using SolidWorks and fabrication processes.

Numerical Analysis of NACA 4312 Airfoil

- Performed CFD analysis using ANSYS Fluent to evaluate lift, drag, and pressure distribution.
- Analyzed velocity contours to study aerodynamic performance under varying conditions.

Gearbox Design for Friction Washer

- Designed gearbox system and performed torque and power transmission calculations.
- Modeled assembly components using Solid Edge.

Reverse Engineering of Bicycle Air Pump

- Reverse engineered a bicycle air pump and developed CAD models of internal components.
- Conducted material identification using FTIR and microscopic analysis.

Smart Waste Management System

- Developed a CNN-based waste classification system using TensorFlow and OpenCV.
- Implemented automated sorting using Raspberry Pi and sensor integration.

SKILLS

Languages: English (professional proficiency), Sinhala (native proficiency)

Software:

- **3D Modeling:** SolidWorks, Solid Edge, AutoCAD, Rhinoceros 3D
- **Simulation:** SolidWorks Simulation, ANSYS, MATLAB (Simulink, Simscape Multibody), Orca3D, Maxsurf

Engineering: Mechanical Design, Design for Manufacturing (DFM), CAD Modeling, CFD Analysis, Structural Analysis, Mechanism Design, Steering System Design, Fabrication and Prototyping

Programming: Python, C++

Soft Skills: Communication, Leadership, Teamwork

HONORS & AWARDS

IMechE SL Group Final Year Project Award

Dec 2025

Institution of Mechanical Engineers (IMechE), Sri Lanka Group

- Awarded for outstanding performance, innovation, and teamwork in final year engineering project.
- Selected among top projects evaluated by IMechE Sri Lanka Group.

REFERENCES

Dr. Lihil Subasinghe

Senior Lecturer, University of Moratuwa

subasinghe@uom.lk

Mr. Chanaka Rathnayaka

CAD/CAM Lead, Vega Innovations (Pvt) Ltd

chanaka@vega.lk